SatensoRF: Fast Satellite Tensorial Radiance Field for Multidate Satellite Imagery of Large Size

Tongtong Zhang, Yu Zhou, Yuanxiang Li, *Member, IEEE*, and Xian Wei, *Senior Member, IEEE*

***Abstract*— Existing neural radiance field (NeRF) models for satellite imagery have limitations in processing large images and require solar input, leading to slow speeds. As a response, we introduce SatensoRF, which speeds up the entire process significantly while using fewer parameters for large satellite imagery. We have noticed that the common assumption of Lambertian surfaces in satellite NeRFs (Sat-NeRFs) is not sufficient for vegetative and aquatic elements. In contrast to the traditional hierarchical multilayer perceptron (MLP)-based scene representation, we have chosen a multiscale tensor decomposition approach for color, volume density, and auxiliary variables to model the light field with specular color. Additionally, to rectify inconsistencies in multidate imagery, we incorporate total variation denoising (TVD) to restore the density tensor field, thus mitigating the negative impact of transient objects. To validate our approach, we conducted assessments of SatensoRF using subsets from the SpaceNet multiview dataset, which includes both multidate and single-date multiview RGB images. Our results demonstrate that SatensoRF surpasses the state-of-the-art Sat-NeRF series regarding novel view synthesis performance. Significantly, SatensoRF requires fewer parameters for training, resulting in faster training and inference speeds and reduced computational demands.**

***Index Terms*— Acceleration, digital surface model (DSM), neural radiance field (NeRF), tensor decomposition.**

I. Introduction

THE reconstruction of the Earth's surface from high-resolution satellite imagery is consistently important across various domains. Conventional approaches in this field typically address the challenge by solving multiple stereo problems to estimate disparity [1], [2], [3], [4], [5]. However, these approaches encounter inherent challenges during the dense matching phase. Despite the integration of deep-learning techniques into satellite imagery, such as the differentiable warping of the rational polynomial camera (RPC) for multiview cost volume in [6], the computational and time costs remain substantial. Besides, most of the deep-learning-based stereo problems require ground-truth depth maps for supervision [7].

In recent years, the emergence of the neural radiance field (NeRF) [8] has sparked a revolutionary shift, particularly within computer graphics and computer vision. Methods for reconstructing 3-D scenes based on NeRF have a significant advantage in that they rely less on dense matching, making a ground-truth depth map unnecessary for supervision. This is particularly useful for reconstructing satellite imagery, where obtaining an accurate and high-resolution depth map can be challenging. Besides, generating high-quality images of a novel view enhances the satellite image data and provides a more complete representation of the 3-D scene [9]. The fundamental concept behind the original NeRF is to use a fully connected neural network to model a continuous scene. This network takes ray samples and viewing directions as inputs and maps them to RGB color and volume density. Subsequently, differential volume rendering facilitates end-to-end training for scene representation. Subsequent adaptations of NeRF in the aerospace domain have produced significant benefits for drone applications [10], [11], as well as satellites [9], [12]. The common domain gap for migrating canonical RGB input to satellite images includes the following aspects.

1) Radiometric inconsistencies among images across multiple dates.
2) Discrepancies in the ray modeling between perspective cameras and RPCs, which require distinct sampling strategies.

For the first gap, S-NeRF [12] introduces a remodeled neural volume rendering approach based on solar irradiance and incorporates the solar directions ω_s as input, as depicted in Fig. 1(b). Building upon this, satellite NeRF (Sat-NeRF) [9] extends the framework to model transient objects by regressing uncertainty to adjust the loss dynamically. Sat-NeRF also draws inspiration from NeRF in the wild (NeRF-w) [13], by introducing image index embedding t_j to supplement the multidate difference in Fig. 1(c). For the second gap, Sat-NeRF also adapts the point sampling model to accommodate RPCs. Despite the successful adaptation of NeRFs to multidate satellite imagery, there are still significant practical challenges that persist.

The first left-over problem is that the conventional NeRF parameterization techniques in S-NeRF and Sat-NeRF inherently retain the efficiency limitations of vanilla NeRF. The

Manuscript received 18 September 2023; revised 11 January 2024 and 23 February 2024; accepted 19 March 2024. Date of publication 27 March 2024; date of current version 19 April 2024. This work was supported in part by the Special Funds for Creative Research under Grant 2022C61540. *(Corresponding author: Yuanxiang Li.)*

Tongtong Zhang and Yuanxiang Li are with the Department of Information and Control, School of Aeronautics and Astronautics, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: yuanxli@sjtu.edu.cn).

Yu Zhou is with Xi'an Institute of Surveying and Mapping, Xi'an 710054, China.

Xian Wei is with Software Engineering Institute, East China Normal University, Shanghai 200050, China.

Digital Object Identifier 10.1109/TGRS.2024.3382632

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

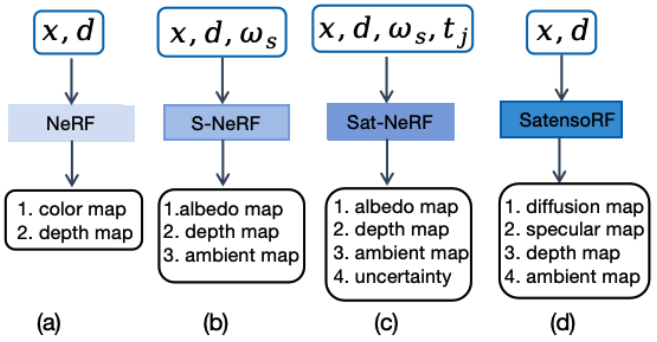


Fig. 1. Flow diagram comparison of SatensoRF and the state-of-the-art satellite-related NeRFs. SatensoRF does not have any additional input requirements other than the position x and viewing direction d. ω_s represents the solar direction, and t_j represents the image index embeddings. (a) NeRF. (b) S-NeRF. (c) Sat-NeRF. (d) SatensoRF.

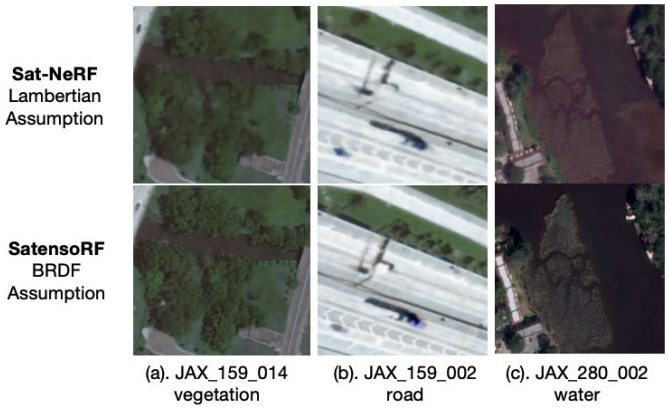


Fig. 2. Rendering results on specific patches with different light field assumptions. (a) For vegetational regions, SatensoRF with BRDF assumptions yields better tree details. (b) For reflective roads, SatensoRF results in finer edges. (c) SatesoRF is more capable of capturing water regions.

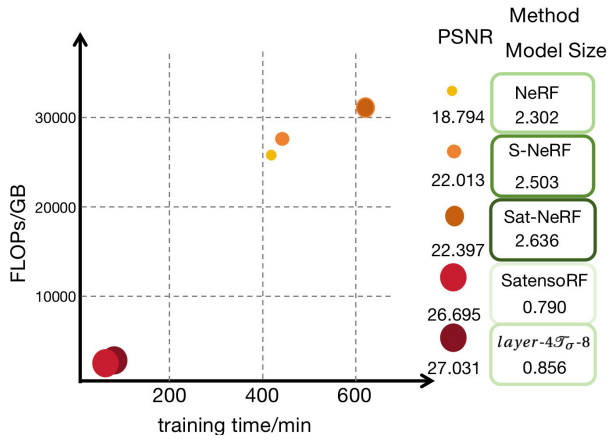


Fig. 3. Comprehensive comparison of SatensoRF with SOTA satellite NeRFs, on subset 159 of the SpaceNet multiview dataset with original size. SatensoRF achieves better rendering results with faster training speed with less computation cost, and layer $-4\mathcal{T}_\sigma-8$ refers to SatensoRF with four layers MLP and eight levels of tensor radiance fields.

two-stage coarse-to-fine training approach relies on having a sufficient number of images from different viewpoints to ensure the synthesis of continuous views. This rendering configuration leads to dense querying of multilayer perceptrons (MLPs) at all query points. Besides, the purely MLP-based model introduces a tradeoff between memory consumption and batch size, consequently impeding the overall efficiency and constraining the input image size.

The second problem is that Sat-NeRF requires additional input such as sun azimuth, and elevation, as shown in Fig. 1. On the one hand, these additional inputs are essential for Sat-NeRF but might be inaccessible for some datasets. On the other hand, additional input tensors cause more computations and further slow the running process.

The third problem is the oversimplified light field assumption. S-NeRF and Sat-NeRF have operated under the assumption that the Earth's surface follows strict Lambertian law. The rendered color is broken down into distinct elements, such as surface albedo, ambient color, and adjusting factor, which depend on solar orientation. Real-world surfaces often deviate from Lambertian qualities and exhibit varying degrees of bidirectional reflection traits [14]. This phenomenon applies to surfaces such as bodies of water, vegetation, and asphalt roads, and the distinction is illustrated in Fig. 2.

This article presents SatensoRF, a solution that tackles three challenges with a unified architecture. The first efficiency challenge involves model design. For the second challenge, two specific additional inputs should be removed, namely, the solar direction and the index embedding. And for the third challenge, we introduce a revised light field model with SatensoRF in Section III-A to account for the bidirectional reflectance distribution function (BRDF) properties. Therefore, we tackle the three challenges together with the unified design of SatensoRF.

1) SatensoRF addresses the efficiency challenge by using a level-of-detail (LoD) representation to tensorize the 3-D scene. This leads to a considerable improvement in running speed and a reduction in the overall model size.
2) Instead of relying on the solar direction to create a light field on a Lambertian surface as [9], [12], SatensoRF learns the light decomposition model, modeling the sun's effect via specular light, eliminating the need for additional solar information.
3) Rather than encoding the transient objectives in multidate photo collections via image index embedding in [12], [13], and [9], SatensoRF fully exploits the low-rank property of satellite images. The main component of the entire scene is considered to be static and parameterized by low-rank tensors, while the density of the transient objects is regarded as noise to be eliminated. Thus, mitigating the influence of temporary objects on density estimation is translated into denoising the stable volume density field, as further elucidated in Section III-D.

For the first challenge, SatensoRF requires fewer parameters, which means it requires less memory and is more suitable for processing large satellite images compared to Sat-NeRF. For the second and the third challenges, in Fig. 2, the proposed model encompassing the reflective light yields finer details of trees, the metal surface of cars, and the water surface.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

To sum up, this article makes several contributions by addressing the three challenges with one unified model.

1) Introducing a lightweight and accelerated tensor radiance field framework that minimizes training and inference time.
2) Reducing model sizes and overall computational memory costs, while demonstrating superior results on large-sized satellite images compared to state-of-the-art NeRF-based approaches, as Fig. 3 shows.
3) Capturing the natural anisotropic scattering properties of the Earth's surface without assuming the Lambertian law is accomplished through a lightweight model design that offers enhanced performance.
4) Removing input dependencies by enabling SatensoRF to operate without relying on solar direction information.

The proposed SatensoRF is evaluated on different areas of interest, the multidate WorldView-3 images of the original size. For altitude estimation, we adopt a LIDAR digital surface model (DSM) of resolution 0.5 m/pixel as an evaluation reference. SatensoRF is compared to the state-of-the-art NeRF, S-NeRF, and Sat-NeRF.

II. Related Work

A. Acceleration Framework for NeRFs

In the conventional NeRF approach, the rendering step involves the dense evaluation of MLPs at all query points along each camera ray during numerical integration, leading to slow speed and extensive computational demands. To address this, baking techniques have been introduced to accelerate inference time [15], [16], [17]. In pursuit of swifter training and inference, the utilization of diverse formats for learnable features yields more concise outcomes. While MLPs exhibit competitiveness in neural implicit representations [18] and original NeRFs, they lag behind methods that directly optimize and query the scene representations themselves in terms of speed. Notably, Instant-NGP [19] employs multiresolution hash tables for scene representation organization, achieving fast processing with a level of detail. Other structures like trees [20] and sparse grids [16], [21] are also popular. In parallel, methods such as [22], [23], [24] structure the scene as tensor compositions with low rank. TensoRF [22], for instance, employs vector–matrix (VM) and CANDECOMP/PARAFAC (CP) decomposition to characterize the scene, followed by input to compact MLPs for efficient computation. Similarly, CCNeRF [23] discards the use of MLPs to facilitate compression and composition. NRFF extends TensoRF into a multiscale framework to enable comprehensive learning at varying scales, thereby enhancing accuracy. Coincidentally, remotely sensed images often exhibit low-rank properties [25], [26], which can be further exploited in the restoration task. The proposed SatensoRF makes full use of the TensoRF design and tailors the multiscale version of TensoRF to address the unique challenges posed by satellite images.

B. NeRF With Multidate Images

The vanilla NeRF model conditions the fact that the 3-D scene has density, radiance, and illumination, which is unreal for outdoor scenes, especially for internet photo collections and satellite images. Both of these two types of datasets have different lighting conditions and the appearance of the pixel at the same 3-D position. Besides, there are transient occlusions from people and cars. NeRF-W [13] first proposes a per-image latent embedding capturing photometric appearance variations often present in wild heritage datasets and explicitly models the scene into image-dependent and shared components, respectively, to disentangle transient elements from the static scene. With the same starting point, Sat-NeRF adapts the method to satellite images by representing the albedo of the Earth's surface, and the other lighting factors in different heads of MLPs. NeRF-OSR [27] enables outdoor scene relighting.

C. Deep Learning in Reconstruction From Satellite Images

Traditional 3-D reconstruction has predominantly followed a multiview stereo (MVS) pipeline that generates a dense disparity map using dense matching algorithms such as semi-global matching (SGM) [28]. However, errors tend to accumulate from the initial stereo pair selection to the subsequent refinement steps [28], [29]. As deep-learning methodologies infiltrate the realm of 3-D reconstruction, an increasing number of end-to-end differentiable MVS pipelines have emerged [30]. Some approaches extract features from the 3-D cost volume and utilize 3-D convolutional neural networks (CNNs) [31], [32], recurrent neural networks (RNNs) [33], or attention modules [34] to predict the disparity map. However, these methods often require significant computation and time. On the other hand, some methods adapt randomized iterative patch-match techniques [35], [36] into an end-to-end framework [37], [38].

The proliferation of implicit neural representations (INRs) introduces increased flexibility in terms of computation and memory utilization. Traditional epipolar geometry and INR complement each other in the realm of 3-D reconstruction. MVS provides geometric guidance for the radiance field [39], [40], [41], while INR reduces the reliance on conventional MVS pipelines for selecting optimal stereo pairs and explicit depth supervision [42], [43].

III. Method

SatensoRF addresses the challenges of efficiency, additional output, and new light field assumptions via tensorizing the NeRF. The motivation of the light field decomposition is recorded in Section III-A. The whole pipeline is given in Fig. 4, with Section III-B explaining how to implement the light field decomposition within the pipeline. The submodule reflective field is discussed in Section III-C, and the density field denoising against transient objects is further discussed in Section III-D.

A. Light Field Decomposition

This section explains how to model the sun effect without relying on solar direction information, by decomposing the light field. The primary objective of SatensoRF is to generate a synthesized novel view photo in RGB, along with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

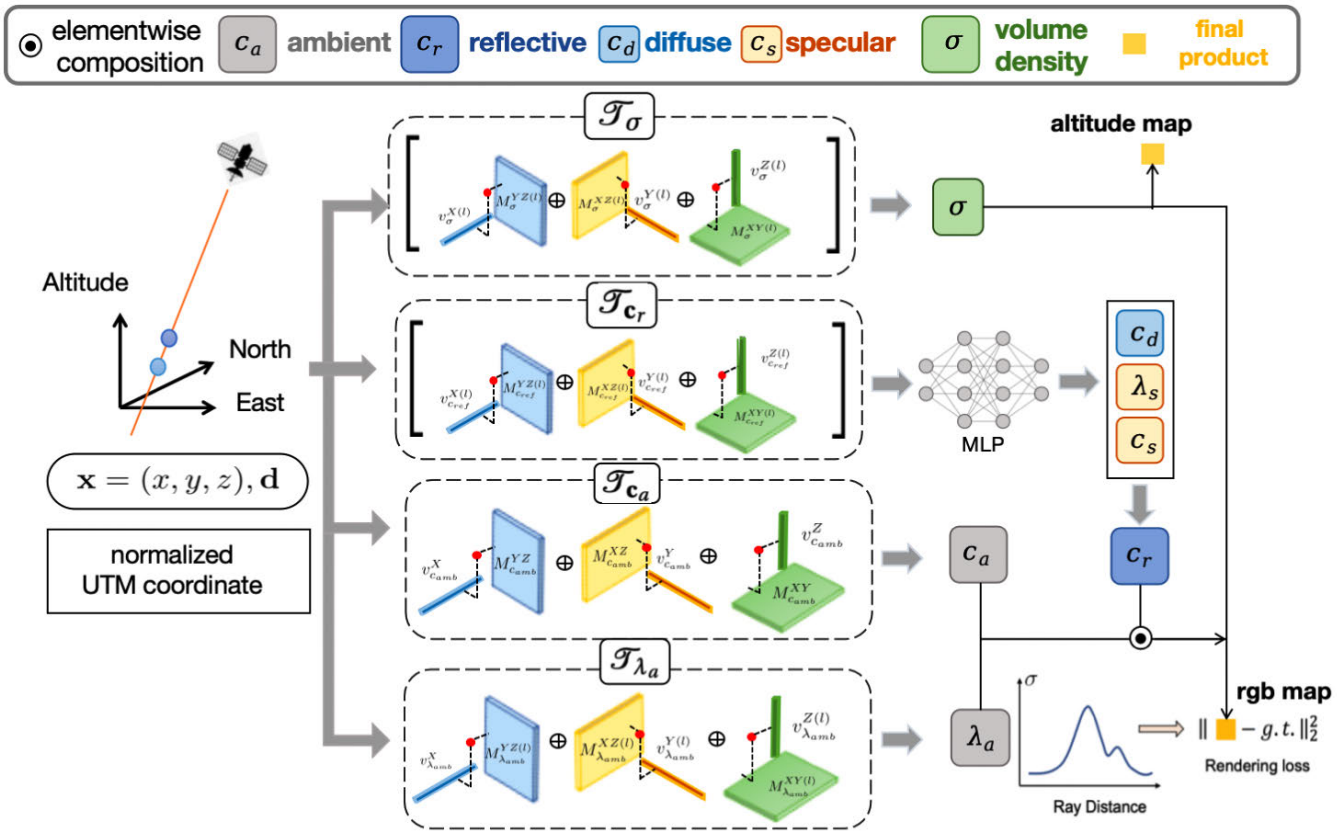


Fig. 4. SatensoRF pipeline, with four distinct subtensor radiance fields: 1) the multiscale volume density field $\mathcal{T}_\sigma$ to estimate the volume density σ, 2) the reflectance field $\mathcal{T}_{\mathbf{c}_r}$ to estimate the reflective color $\mathbf{c}_r$ is fed to the MLPs for further processing, as explained in Section III-C, and 3) both the ambient color field $\mathcal{T}_{\mathbf{c}_a}$ and the ambient factor field $\mathcal{T}_{\lambda_a}$ operate on a single-level scale estimate $\mathbf{c}_a$ and λ_a, respectively. Finally, (3) and (4) are applied for synthesizing image $\hat{I}$ and altitude $\hat{H}$.

an altitude map product. The input of SatensoRF includes the spatial coordinates of the scene volume, and the viewing direction $\mathbf{d}$, as Fig. 1(d). For a sampled point $\mathbf{x}_i = \mathbf{r}(t_i) = \mathbf{o} + t_i\mathbf{d}$, SatensoRF predicts both the rendered color $\mathbf{c}(\mathbf{x}_i, \mathbf{d})$ and the volume density $\sigma(\mathbf{x}_i, \mathbf{d})$ for the ray $\mathbf{r}(t_i)$ at a given sample t_i within the interval $[t_{\min}, t_{\max}]$. Here, $t_{\min}$ and $t_{\max}$ define the sampling range along the altitude direction.

Due to the light field complexity of the Earth's surface, the rendered color $\mathbf{c}$ is calculated as a composition of several color components output by the model. Different from the previous works [9], [12], the reflection of the underlying surface consists of diffuse reflection and specular reflection. For convenience, the notations of the color components involved in the computation are listed as follows.

1) *Ambient Color* $\mathbf{c}_a$*:* The overall hue of the environment (unlike [9], [12], SatensoRF does not assume $\mathbf{c}_a$ to rely on the solar direction, as shown in Fig. 4).
2) *Ambient Color Intensity* λ_a*:* The adjusting factor indicating the intensity of $\mathbf{c}_a$.
3) *Specular Color* $\mathbf{c}_s$*:* The color of highlight when light rays are reflected directly off the Earth's surface without being absorbed or scattered.
4) *Specular Intensity* λ_s*:* The adjusting factor indicating the intensity of the specular reflection $\mathbf{c}_s$.
5) *Diffusion Color* $\mathbf{c}_d$*:* The intrinsic color of the Earth's surface under pure white light only depends on the position $\mathbf{x}_i$ (the incoming light is reflected in all directions equally).

Ultimately, with the design of SatensoRF, for the 3-D point $\mathbf{x}_i$ and the viewing direction $\mathbf{d}$, we achieve the following composition:

$$\begin{aligned} \mathbf{l}(\mathbf{x}_i) &= \lambda_a(\mathbf{x}_i) l_3 + (1 - \lambda_a(\mathbf{x}_i))\mathbf{c}_a(\mathbf{x}_i) \\ \mathbf{c}_r(\mathbf{x}_i, \mathbf{d}) &= \mathbf{c}_d(\mathbf{x}_i) + \lambda_s \mathbf{c}_s(\mathbf{x}_i, \mathbf{d}) \\ \mathbf{c}(\mathbf{x}_i, \mathbf{d}) &= \mathbf{c}_r(\mathbf{x}_i, \mathbf{d}) \odot \mathbf{l}(\mathbf{x}_i) \end{aligned} \tag{1}$$

where l_3 represents the white light source, $\odot$ denotes element-wise multiplication at each position, and λ_a adjusts the impact of the ambient color $\mathbf{c}_a$. The irradiance map $\mathbf{l}(\mathbf{x}_i)$ refers to the effect ambient light map $\mathbf{l}$ at $\mathbf{x}_i$, and it is weighted with both the ambient color $\mathbf{c}_a$ and the white light l_3.

The final color $\mathbf{c}$ is composited of the irradiance map $\mathbf{l}$ and the reflected field $\mathbf{c}_r$, where the reflected radiance field for $\mathbf{c}_r$ will be discussed in detail in Section III-C.

B. Pipeline of SatensoRF

As shown in Fig. 4, the components of the decomposed light field in Section III-A is accomplished with four subfields to access the ambient color $\mathbf{c}_a$, the ambient factor λ_a, the volume density σ and the reflective color $\mathbf{c}_r$ from ground surface

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

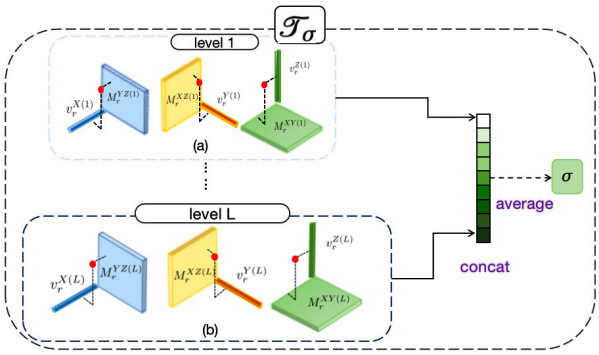


Fig. 5. Multiscale tensor factorization of the volume density field $\mathcal{T}_\sigma$ computed and then averaged to produce the volume density field. (a) Tensor field at level 1 and (b) tensor field at level L.

for (1). Concretely, SatensoRF encompasses subfields with the following notations.

1) *Volume Density Field $\mathcal{T}_\sigma$:* The radiance feature field for volume density. It has $L = 8$ levels of tensor decomposition, and the outputs are concatenated and averaged across levels to yield volume density σ, as shown in Fig. 5.
2) *Reflective Field $\mathcal{T}_{\mathbf{c}_r}$:* The radiance feature field for appearance with output $\mathbf{c}_r$. It also has $L = 8$ levels, and the outputs are concatenated and fed into a small multihead MLP network, with further details provided in Section III-C.
3) *Ambient Color Field $\mathcal{T}_{\mathbf{c}_a}$:* The irradiance feature field for ambient color. It has only one level of tensor decomposition with output $\mathbf{c}_a$.
4) *Ambient Factor Field $\mathcal{T}_{\lambda_a}$:* The irradiance field feature field. It has only one level of tensor decomposition output λ_a.

Considering that $\mathcal{T}_\sigma$ and $\mathcal{T}_{\mathbf{c}_r}$ encompass multiscale details, these two feature fields incorporate multiple levels of tensor decomposition. The coarse-to-fine strategy is a common approach in NeRF pipelines, as discussed in [8]. In SatensoRF, we decompose the feature field $\mathcal{T}_\sigma$ and $\mathbf{c}_r$ for the entire 3-D scene into L levels. The sizes of each level are selected using a strategy akin to NGP [19], based on logarithmic considerations. Take the volume density field $\mathcal{T}_\sigma$, for example, in Fig. 5, there are $L = 8$ levels of tensor factorizations from (a) to (b), respectively.

For each singular level, basic VM decomposition factorizes a tensor into multiple vectors and matrices as TensoRF [22]. The three dimensions of the 3-D space are designated as XYZ, wherein X and Y correspond to the north and east directions in the Universal Transverse Mercator (UTM) coordinate system, while Z signifies the elevation of the Earth's surface. Concretely, VM decomposition for a given single level l can be formulated as follows:

$$\mathcal{T}^{(l)} = \sum_{r=1}^{R} (v_r^{X(l)} \odot M_r^{YZ(l)} + v_r^{Y(l)} \odot M_r^{XZ(l)} + v_r^{Z(l)} \odot M_r^{XY(l)}) \tag{2}$$

where M_r^* are matrix factors for two of the three directions, $\odot$ is the element-wise multiplication, and R is the rank at three direction.

In contrast to $\mathcal{T}_\sigma$ and $\mathcal{T}_{\mathbf{c}_r}$, $\mathcal{T}_{\mathbf{c}_a}$ and $\mathcal{T}_{\lambda_a}$ require fewer parameters and are characterized by a singular scale as (2). Finally, with the predicted $\sigma, \mathbf{c}_r$, the final rendered map $\hat{I}$ with basic volume rendering formulation [8] can be given by

$$\begin{aligned} \hat{I} &= \sum_{i=1}^{N_s} \mathrm{Tr}(\mathbf{x}_i)\alpha(\mathbf{x}_i)\mathbf{c}(\mathbf{x}_i, \mathbf{d}) \\ &= \sum_{i=1}^{N_s} \mathrm{Tr}(\mathbf{x}_i)\alpha(\mathbf{x}_i)\mathbf{c}_r(\mathbf{x}_i, \mathbf{d})\mathbf{l}(\mathbf{x}_i) \end{aligned} \tag{3}$$

where the opacity $\alpha(\mathbf{x}_i)$ and transmittance $\mathrm{Tr}(\mathbf{x}_i)$ are calculated as

$$\alpha(\mathbf{x}_i) = 1 - \exp(-\sigma(\mathbf{x}_i)\delta_i) \tag{4}$$

$$\mathrm{Tr}(\mathbf{x}_i) = \prod_{j=1}^{i-1}(1 - \alpha(\mathbf{x}_j)) \tag{5}$$

with δ_i being the two consecutive points along the ray, that is, $\delta_i = t_{i+1} - t_i$, and N_s is the number of samples in the altitude direction. The ray depth follows with the same integration, and the estimated altitude in the normalized coordinate eliminates the impact of the ray direction

$$\begin{aligned} \hat{D} &= \sum_{i=1}^{N_s} \mathrm{Tr}(\mathbf{x}_i)\alpha(\mathbf{x}_i)t_i \\ \hat{H} &= \hat{D} + (1 - \sum_{i=1}^{N_s} \mathrm{Tr}(\mathbf{x}_i)\alpha(\mathbf{x}_i)) \cdot \mathbf{d}. \end{aligned} \tag{6}$$

C. Reflective Radiance Field $\mathcal{T}_{\mathbf{c}_r}$

The reflected color $\mathbf{c}_r$ of can be decomposed into diffusion color $\mathbf{c}_d$ and specular light $\mathbf{c}_s$ with factor λ_s [44], [45]

$$\begin{aligned} \mathbf{c}_r(\omega_o; \mathbf{x}) &= \mathbf{c}_d + \lambda_s \mathbf{c}_s \\ &= \mathbf{c}_d + \lambda_s \int_\omega L_i(\omega_i; \mathbf{x})\rho_s(\omega_i, \omega_o; \mathbf{x})(\mathbf{n} \cdot \omega_i) d\omega_i \\ &= \mathbf{c}_d + \lambda_s \int_\omega f(\omega_i, \omega_o; \mathbf{x}, \mathbf{n}) d\omega_i \end{aligned} \tag{7}$$

where L_i indicates the incoming radiance from direction ω_i around the surface normal $\mathbf{n}$ at position $\mathbf{x}$, ω_o is the outgoing irradiance direction, ρ_s is the BRDF depending on $\mathbf{x}$ and ω_i, and the dot product between ω_i and $\mathbf{n}$ serves as the cosine of the intersecting angle. The specular part is the integration over the hemisphere Ω centered around $\mathbf{n}$ containing all possible values for ω_i where $\omega_i \cdot \mathbf{n} > 0$. Instead of solving the rendering equation via Monte Carlo estimation, implicit representations learn the components of the rendering equation via neural networks [24], [46].

Subsequently, we perform a decomposition of the reflective radiance field using the ambient color feature field $\mathcal{T}_{\mathbf{c}_a}$. The extracted features are then fed to MLPs. According to (8), both the diffusion color $\mathbf{c}_d$ and the factor λ_s remain independent of the viewing direction. As a result, they can be directly regressed through MLPs. In this case, a four-layer MLP structure (labeled as "A" in Fig. 6) is employed to predict the diffusion color $\mathbf{c}_d$, the factor λ_s, and the normals n for every volume intersected by the sampled ray.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

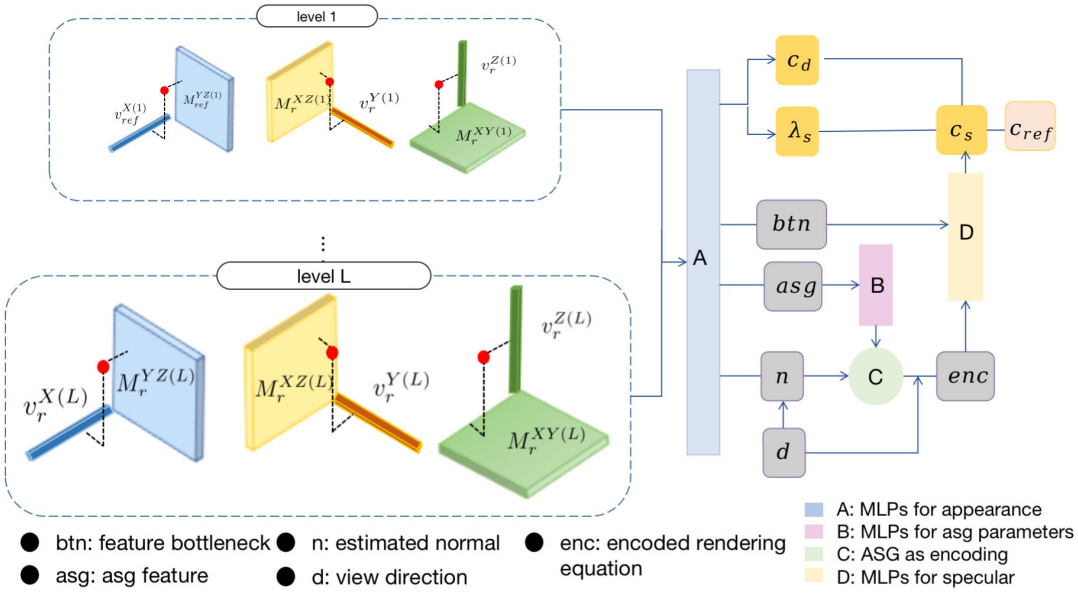


Fig. 6. Multihead MLPs to decouple the reflective light field $\mathcal{T}_{\mathbf{c}_r}$.

To improve the parameterization of the rendering equation, the encoding process is applied within the color feature space, which differs from most NeRF methods that perform encoding in the scene coordinate space [8], [24]. This approach is particularly advantageous because the BRDF function f depends solely on the specific features of the color domain. The encoded feature of the rendering equation is modeled as an anisotropic spherical Gaussian (ASG) function, a concept introduced by the work [47]. This ASG function is composed of subfunctions G_i that further enhance the ability to represent intricate scene characteristics

$$\mathcal{F}_s(\omega_o, \mathbf{x}) = \sum_{i=0}^{N-1} G_i(\omega_o; \mathbf{x}, [\omega_i, \omega_i^{\lambda}, \omega_i^{\mu}], [\lambda_i, \mu_i], a_i) \tag{8}$$

$$= \sum_{i=0}^{N-1} F_i \max(\omega_o \cdot \omega_i, 0) \exp(-\lambda_i(\omega_o \cdot \omega_i^{\lambda})^2 - \mu_i(\omega_o \cdot \omega_i^{\mu})^2)$$

where F_i is the feature vector, $\mathcal{B} = [\omega_i, \omega_i^{\lambda}, \omega_i, \omega_i^{\mu}]$ are orthonormal basis, with $\lambda_i, \mu_i > 0$ being the bandwidth at each basis.

The parameters of the ASG F_i, λ_i, and μ_i are initially obtained from two MLPs. The first MLP labeled "A" (blue) and another two-layer MLP labeled "B" (pink) in Fig. 6 contribute to determining these ASG parameters. The basis $\mathcal{B}$ required for ASG representations is sampled on a unit sphere. These basis vectors are then processed through the encoder labeled "C" (green) in Fig. 6. The encoder generates the rendering equation that is contingent on the viewing direction, together with the intersection angle derived from the dot product of the normals n and the viewing direction d. Subsequently, the specular part $\mathbf{c}_s$ is decoded using the last two-layer MLP labeled "D" in Fig. 6, employing a Sigmoid activation function. This decoded specular part $\mathbf{c}_s$ then contributes to forming the reflective color $\mathbf{c}_r$.

D. Density Field Restoration via Total Variation Denoising (TVD)

To avoid the need for additional input such as image index embedding, we use an alternative strategy to account for transient objects without regressing the uncertainty in [9]. Specifically, we take full advantage of the low-rank property of SatensoRF using a standard tensor denoising technique.

The principal assumption made here is that the scene can be represented by compositing low-rank tensors, which bear resemblance to the assumptions of low-rank image denoising [48]. On the one hand, TVD is a widely employed method for signal restoration tasks. This regularization technique is particularly effective in preserving sharp edges present within the underlying signal [49], [50]. On the other hand, the total variation loss serves as a prevalent regularization technique in compressed sensing, a requirement innate to the proposed architecture.

In SatensoRF, the Earth's surface is considered as the signal to be restored from the noise-contaminated input data, represented by the volume density tensor $\mathcal{X}\sigma$ composed of the model parameters of the volume density field $\mathcal{T}\sigma$. Simultaneously, transient objects such as cars or people exhibit discernible patterns and infrequent occurrences compared to the overall static scene. These unwanted patterns, resembling noise, are considered as a sparse noise tensor $\mathcal{N}$ in this context.

Therefore, we denote the true volume density tensor as $\mathcal{T}_\sigma$ and subsequently formulate the volume density tensor as follows:

$$\mathcal{T}_\sigma = \mathcal{X}_\sigma + \mathcal{N}. \tag{9}$$

In consideration of this, we apply the concept of total variation loss to the decomposed planes of $\mathcal{T}_\sigma$ across all L levels, to restore the coherent structures of the desired clean

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

volume density field $\mathcal{T}_\sigma$

$$\mathcal{L}_{\text{TV}}(\mathcal{T}_\sigma) = \sum_{l=1}^{L} \|\mathbf{D}\mathcal{T}_\sigma\|_2/L \tag{10}$$

$$\|\mathbf{D}\mathcal{T}_\sigma^{(l)}\|_2 = \sum_{l=1}^{L}\Big(\sum_i (M_\sigma^{XY(l)}(i+1,:) - M_\sigma^{XY(l)}(i,:)) + \sum_j (M_\sigma^{XY(l)}(:,j+1) - M_\sigma^{XY(l)}(:,j))\Big) \tag{11}$$

where $\mathbf{D}$ is the difference operator and i, j index elements on the factorized matrix $M_\sigma^{XY(l)}$ at scale l from the L levels.

E. Losses

1) Photometric Loss $\mathcal{L}_{rgb}$: The basic supervision is the mean squared error (MSE) between the rendered color at every ray $\mathbf{r}$ and the ground-truth color of the input images

$$\mathcal{L}_{\text{rgb}} = \sum_{\mathbf{r}\in\mathbf{R}} \|\hat{\mathbf{c}}(\mathbf{r}) - c_{gt}((\mathbf{r})\|_2^2 \tag{12}$$

2) Normalized Surface Loss $\mathcal{L}_{normal}$: Since all the rays are normalized, all the sampled points are assumed inside the unit cube, and each ray direction $\mathbf{d}$ is also normalized. To encourage all the normalized surface normals $\mathbf{n}_i, i \in \{1,\ldots,N-1\}$ to be above the Earth's surface that contribute to the ray's rendered color [51], the normal loss penalizes samples with high volume density ω_i and invalid normals with $\mathbf{d}\cdot\mathbf{n}_i > 0$

$$\mathcal{L}_{\text{normal}} = \frac{1}{N}\sum_{i=0}^{N-1} \omega_i \max(0, \mathbf{d}\cdot\mathbf{n}_i)^2 \tag{13}$$

where N is the number of samples in the cast ray. $\mathcal{L}_{\text{normal}}$ indicates that the volumes with invalid normals and high volume density are discouraged.

3) Ray Stop Regularization $\mathcal{L}_{\lambda_a}$: The ambient color $\mathbf{c}_a$ takes no effect when the ray intersects with the Earth's surface, thus we encourage the ratio of the ambient color intensity λ_a to be close to the transmittance Tr

$$\mathcal{L}_{\lambda_a} = \sum_{i=1}^{N_{\text{all}}} (\text{Tr}(\mathbf{x}_i) - \lambda_a(\mathbf{x}_i))^2 + 1 - \sum_{i=1}^{N_{\text{all}}} \text{Tr}(\mathbf{x}_i)\alpha(\mathbf{x}_i)\lambda_a(\mathbf{x}_i) \tag{14}$$

where N_{all} is the number of total voxels.

4) Total Loss: The basic whole loss include the data fidelity term $\mathcal{L}_{\text{rgb}}$, the noise sparse prior $\mathcal{L}_{\text{TV}}$, the normal regularization $\mathcal{L}_{\text{normal}}$, and the ambient color regularization $\mathcal{L}_{\lambda_a}$

$$\mathcal{L} = \alpha\mathcal{L}_{\text{rgb}}(\mathbf{c}, c_{gt}) + \beta\mathcal{L}_{\text{TV}}(\mathcal{T}_\sigma) + \gamma\mathcal{L}_{\text{normal}} + \epsilon\mathcal{L}_{\lambda_a} \tag{15}$$

where $\alpha = 1.0, \beta = 1.0, \gamma = 0.1$, and $\epsilon = 0.1$ indicate the loss coefficients for the losses, respectively.

IV. Experiments

We assess the performance of SatensoRF using subsets of the SpaceNet MVS dataset.[1] The evaluation is conducted from the following aspects.

1) To validate the efficiency of SatensoRF for both single-date and multidate satellite images of large size, a comparison is included in Section IV-B.
2) To validate the effectiveness of SatensoRF without additional input, we compare it to the state-of-the-art Sat-NeRF models (with extra input) for both single-date and multidate satellite images of large size in Section IV-C.
3) To assess the validity of the non-Lambertian assumption for SatensoRF, Section IV-D documents the ablation study.
4) To validate the rationality of the noise decomposition assumption in (9) and the TVD model, an ablation study on the total variation loss for SatensoRF is conducted in Section IV-E.
5) We conduct controlled experiments to investigate the impact of altering certain variables in Section IV-F.

A. Setup

1) Dataset Preparation: Our goal is to test SatensoRF's effectiveness for large multidate images without additional information. We use the original dataset from the Maxar WorldView-3 satellite which captured Jacksonville, Florida, between 2014 and 2016. Unlike S-NeRF and Sat-NeRF, we use full-size RGB images with over 4 million pixels each instead of cropping images into smaller patches based on areas of interest.

The original dataset[2] comprises multidate imagery captured under various lighting conditions, and specific details can be found in Table I. The original multidate dataset contains approximately ten times as many rays as the DFC dataset, with challenging scenarios such as Subset 031, which has a mix of vegetation, buildings, and asphalt roads, Subset 159 with a reflective viaduct, and Subsets 260 and 280 containing bodies of water.

Notably, not all the subsets of Jacksonville have corresponding multispectral images (MSIs) to offer the extra information about sun elevation and sun azimuth, so we pick up two subsets with extra sun information (017 and 159), and two of the subsets (260 and 280) without other clues. Specifically, the extra information of solar elevation and solar azimuth serves as the input of sun position in the compared Sat-NeRF models. For subsets 017 and 159, we take this extra sun information as Sat-NeRF input. For subsets 260 and 280, we take the average 45° as the representation of azimuth ranging from (0°, 90°), and we take the average 90° as the representation of elevation angle ranging from (0°, 180°) as the sun position placeholder for the Sat-NeRF model.

Subsets 031 and 159 are used to validate the accuracy of SatensoRF with the extra accurate solar directions, while subsets 260 and 280 are used to test their functionality with data without solar information. To ensure computation fairness, we maintain consistency by keeping the input the same for subsets 260 and 280. For all the datasets, the compared Sat-NeRF shares the same settings with SatensoRF.

[1] https://spacenet.ai/iarpa-multi-view-stereo-3d-mapping/

[2] https://spacenet.ai/iarpa-multi-view-stereo-3d-mapping/

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

TABLE I

DATASET COMPARISONS. ON THE MVS DATASET FROM SPACENET, WE CREATE TWO TYPES OF DATASETS TO STUDY HOW SATENSORF ACTS ON SINGLE-DATE IMAGERY AND MULTIDATE IMAGERY. # INDICATES THE COUNT OF NUMBERS, AND THE MEASURE OF OVERLAP IS PIXEL. FOR THE SINGLE-DATE DATASET, WE CROP THE RGB IMAGE OF INDEX 006 INTO 512 × 512 PATCHES, AND FOR THE MULTIDATE DATASET, WE USE THE IMAGES OF THE ORIGINAL SIZES. ✓INDICATES THE EXISTENCE OF THE ELEVATION AND AZIMUTH ANGLE OF THE SUN POSITION. ALT. BOUNDS INDICATE THE ALTITUDE RANGE OF THE AREA WITHIN THE IMAGE

	subset	Resolution	#Rays	#Images	overlap[pix]	#train	#test	elevation	azimuth	Alt. bounds [m]
single -date	017-6	512 × 512	262604	41	0.7813	18	2	✓	✓	[-23, 4]
	033-6	512 × 512	262604	41	0.6836	18	2	✓	✓	[-27, 3]
	072-6	512 × 512	262604	41	0.5859	18	2	✓	✓	[-22, 1]
	251-6	512 × 512	262604	41	0.4883	18	2	✗	✗	[-28, 14]
multi -date	031	2400 × 2400	5760000	10	-	11	1	✓	✓	[-23, 4]
	159	2400 × 2400	5760000	11	-	10	1	✓	✓	[-29, 15]
	260	2400 × 2400	5760000	16	-	14	2	✗	✗	[-30, 13]
	280	2400 × 2400	5760000	24	-	21	3	✗	✗	[-29, 3]
DFC	004	780 × 765	596276	11	-	9	2	✓	✓	[-24, 1]
	068	795 × 771	613480	19	-	17	2	✓	✓	[-27, 30]
	214	790 × 761	601292	24	-	21	3	✓	✓	[-29, 73]
	260	794 × 771	612412	17	-	15	2	✓	✓	[-30, 13]

Moreover, we also crop each subset into overlapped multiview imagery of size 512 × 512, to evaluate model performance on the same date. For every chosen original subset of the SpaceNet dataset, we crop the image indexed by 006 in the subset, namely 017-6, 033-6, 072-6, and 251-6. The cropping mode is shown in Fig. 7. For single-date imagery, we also use a subset 251-6 without extra sun information.

2) Implemented Detail: All the SatensoRF and its variants are implemented in PyTorch. For the baseline model, we set $L = 16$ levels of tensor decomposition for both the appearance field $\mathcal{T}_{\text{ref}}$ and the density field $\mathcal{T}_\sigma$, $L = 1$ level for the sky color field $\mathcal{T}_\sigma$. The coefficient for the RGB loss $\mathcal{L}_{\text{rgb}}$ and the total variation loss $\mathcal{L}_{\text{TV}}$ are both 1.0. All the learnable parameters are optimized with the Adam optimizer with a learning rate of 0.0002 for the tensor structure and 0.0001 for the MLPs. Both the learning rates degrade log-linearly to 0.1× to their values in the latest steps. The batch size is set to 8192, and all the experiments are conducted on a single card of NVIDIA GeForce RTX 3090 GPU with 16 GB RAM.

3) Metrics: All the performances are measured with standard image quality assessments which are widely adopted for NeRFs [8].

1) *Peak Signal-to-Noise Ratio (PSNR):* A metric to assess the fidelity of the synthesized image.
2) *Structural Similarity Index (SSIM):* A perceptual image quality metric that quantifies the similarity between the ground truth and the synthesized image.
3) *Learned Perceptual Image Patch Similarity (LPIPS):* A deep-learning-based metric designed for image quality assessment, and here we adopt the trained VGG network.

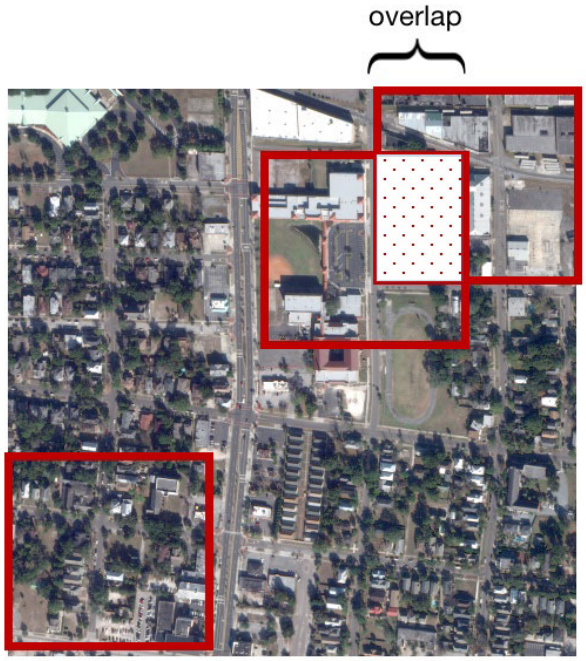


Fig. 7. Illustration of cropping single-date imagery with a fixed size of overlapping to support rational scene geometry for NeRF-based method training.

B. Efficiency Comparison

First, we show that SatensoRF precedes other Sat-NeRFs in efficiency in both the training and inference phases. For multidate datasets, we stop training when the validation PSNR starts to fall, and all of the time is taken to average over different subsets of the dataset. For the single-date dataset, since PSNR increases fast, we stop training after the validation PSNR reaches 36 dB. Here we take the measures below to show the computation efficiency.

1) Training time $T_{\text{train}}^{\text{multi}}(s)$ on the multidate dataset in seconds, until the validation PSNR starts to fall.
2) Training time $T_{\text{train}}^{\text{single}}(s)$ on the multidate dataset in seconds, until the validation PSNR reaches 36 dB.
3) Inference time $T_{\text{test}}^{\text{multi}}(s)$ of computing a total image from the multidate dataset in seconds.
4) Inference time $T_{\text{test}}^{\text{single}}(s)$ of computing a total image from the single-date dataset in seconds.
5) Maximal setting of batch size $BS_{\max}$ for the model to reach the maximum of the single-card GPU memory.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

TABLE II

EFFICIENCY COMPARISON OF SATENSORF AND THE STATE-OF-THE-ART METHODS. *M* IS THE ABBREVIATION FOR MILLION, AND *B* IS THE ABBREVIATION FOR BILLION

	$T_{train}^{multi}(s)$	$T_{train}^{single}(s)$	$T_{test}^{multi}(s)$	$T_{test}^{single}(s)$	BS_{max}	$\#Params(M)$	$FLOPs^{multi}(B)$	$FLOPs^{single}(B)$
NeRF	26648.04	14481.13	131.47	17.23	1024	2.30	26462.45	1204.34
S-NeRF	28021.71	12280.29	136.76	18.02	1024	2.50	28768.67	1309.30
Sat-NeRF	37129.39	13826.52	141.27	19.31	1024	2.64	30293.36	1378.68
SatensoRF	**10129.81**	**2069.65**	**97.49**	**9.35**	**8096**	**0.79**	**9145.22**	**416.21**

TABLE III

NUMERICAL RESULTS COMPARISONS OF SATENSORF AND THE SOTA WORK ON MULTIDATE IMAGES. POSTFIX -SC REFERS TO MODELS WITH SOLAR CORRECTION SUPERVISION IN SAT-NERF PAPER [9] AND -DS REFERS TO MODELS WITH DEPTH SUPERVISION

	Test PSNR$(dB)\uparrow$				**Test SSIM**$\uparrow$				**Test Altitude MAE**$(m)\downarrow$			
Area Name	031	159	260	280	031	159	260	280	031	159	260	280
NeRF	19.595	18.794	16.876	18.921	0.716	0.761	0.483	0.703	3.449	2.628	-	-
S-NeRF + SC	21.059	22.013	17.036	20.514	0.787	0.798	0.504	0.711	2.848	2.318	-	-
Sat-NeRF + SC	21.090	22.397	20.710	20.989	0.792	0.891	0.770	0.783	2.722	1.861	-	-
Sat-NeRF + DS + SC	21.617	23.812	21.094	21.322	0.801	0.901	0.792	0.813	**1.982**	**1.351**	-	-
SatensoRF + SC	24.629	26.695	22.706	24.578	0.921	0.933	0.918	0.939	2.943	2.081	-	-
SatensoRF + DS + SC	**24.771**	**27.013**	**23.151**	**24.866**	**0.934**	**0.929**	**0.952**	**0.931**	1.998	1.859	-	-

TABLE IV

NUMERICAL RESULTS COMPARISONS OF SATENSORF AND THE SOTA WORKS ON THE SEEN CASES OF MULTIDATE IMAGERY. POSTFIX -SC REFERS TO MODELS WITH SOLAR CORRECTION SUPERVISION IN SAT-NERF PAPER [9] AND -DS REFERS TO MODELS WITH DEPTH SUPERVISION

	Train PSNR$(dB)\uparrow$				**Train SSIM**$\uparrow$				**Train Altitude MAE**(m) $\downarrow$			
Area Name	031	159	260	280	031	159	260	280	031	159	260	280
NeRF	26.342	23.847	27.601	24.905	0.723	0.805	0.753	0.823	3.357	2.051	-	-
S-NeRF + SC	22.901	26.088	18.914	21.018	0.799	0.802	0.646	0.724	2.314	2.009	-	-
Sat-NeRF + SC	23.909	24.105	23.782	21.972	0.835	0.914	0.881	0.831	2.531	1.779	-	-
Sat-NeRF + DS + SC	24.101	24.597	24.710	22.849	0.821	0.909	0.804	0.821	**1.759**	**1.104**	-	-
SatensoRF + SC	29.639	32.705	30.988	29.618	0.932	**0.958**	0.924	**0.951**	2.815	2.019	-	-
SatensoRF + DS + SC	**29.872**	**33.382**	**31.210**	**29.905**	**0.956**	0.947	**0.962**	0.949	1.762	1.596	-	-

6) Number of parameters #Params: number of the total parameters with *M* indicating a million.
7) FLOPs^{multi} (floating point operations) for multidate imagery with *B* indicating a billion.
8) FLOPs^{single} (floating-point operations) for single-date imagery with *B* indicating a billion.

The basic comparisons are listed in Table II. As can be seen, the full-size SatensoRF model has roughly a quarter of the parameters of Sat-NeRF models, with the memory cost being far less during training and testing, so the batch size of SatensoRF can be set to nearly 8×–16× of the Sat-NeRF batch size. Besides, the computation of SatensoRF is roughly one-third of Sat-NeRF measured in FLOPs.

C. Performance Comparison

The output of SatensoRF is twofold, a synthesized novel view and an altitude map, so we compare SatensoRF to the extant NeRFs for satellite with satellite cameras. The comparisons mainly include vanilla NeRF [8], S-NeRF [12], and Sat-NeRF [9]. Since the experiments in [9] have validated the effectiveness of solar correction loss for Sat-NeRF, we only take Sat-NeRF with solar correction (SC) for comparison to yield better results, and the notations are clarified as follows.

1) *S-NeRF + SC:* S-NeRF with solar correction.
2) *Sat-NeRF + SC:* Sat-NeRF with solar correction.
3) *Sat-NeRF + DS:* Sat-NeRF with depth supervision.
4) *Sat-NeRF + DS + SC:* Sat-NeRF with solar correction and depth supervision.
5) *SatensoRF + SC:* SatensoRF with solar correction.

1) Multidate Comparison: Table III presents the multidate comparison of SatensoRF and the SOTA NeRF works for multidate imagery on test images unseen by the trained model. With shorter training time and inference time, SatensoRF achieves better results in novel view synthesis tasks with PSNR and SSIM as metrics, and the quality of the reconstructed DSM products equals Sat-NeRF by and large. The MAE of subsets 260 and 280 are omitted because there is water within the area, and we did not perform exact segmentations to find the accurate areas to compute DSM. Table IV presents the multidate comparison for multidate imagery on the training images. SatensoRF converges faster to higher PSNR than

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

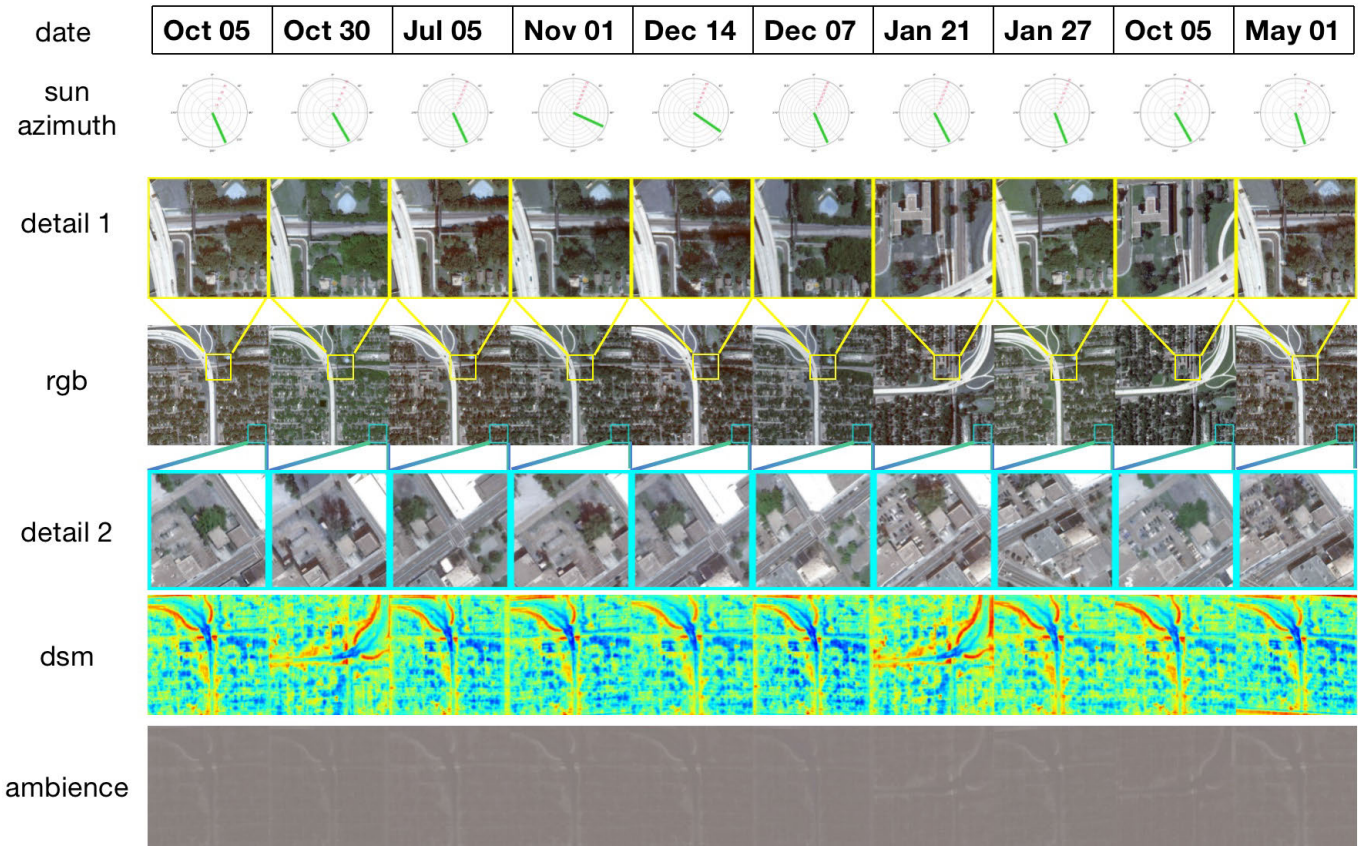


Fig. 8. Details of the synthesized images in subset 031, street details of size 500 × 500, and vegetation details of size 300 × 300, with accurate solar direction as input.

all the NeRF-based models, and the SSIM has increased more during shorter training time, synchronously with PSNR.

For subsets 031 and 159 with accurate solar direction, the synthesized results are presented in Figs. 8 and 9, respectively. The first and second rows show the different dates and the sun azimuth of the first ten images synthesized by SatensoRF. The third and fifth rows show two levels of detail in the synthesized images. The sixth row shows the estimated altitude map from the synthesized DSM, while the last row shows the ambient map (c_{amb}) of different dates. With the multidate images of the original large size, SatensoRF performs better than Sat-NeRF and previous methods.

Meanwhile, the results of images without solar information are presented in subsets 260 in Fig. 10 and 280 in Fig. 11, respectively. This leads to the conclusion that SatensoRF performs better than Sat-NeRF and previous methods, even without solar information.

2) Single-Date Comparison: With the same comparison settings, a comparative analysis between SatensoRF and state-of-the-art NeRF models has been conducted for single-date imagery, as documented in Table V. The evaluations were performed on test and training images from a nearby area. The training was stopped once the validation PSNR reached a threshold of 37 dB. As a result, all models demonstrate comparable performance on the test set, while showing slight differences on the training set. It is worth noting that, under identical lighting conditions, the additional modules of S-NeRF and Sat-NeRF do not promote synthesis quality more than NeRF, whereas vanilla NeRF is faster. Additionally, SatensoRF converges faster than all NeRF-based models with similar performance levels.

D. Ablation Study I: Non-Lambertian Assumption

This section aims to validate the non-Lambertian assumption by modeling the anisotropic reflectance field. The candidates for comparison are as follows.

1) *Lamb CP + SH:* The Lambertian version of SatensoRF without modeling the reflective light. The network has only the diffusion head (MLPs of four layers with 128 hidden neurons), and no other output from module A in Fig. 6. The final RGB output is rendered with spherical harmonics (SHs) functions, and the tensor factorization adopts CP decomposition.
2) *Non-Lamb CP + ASG:* The non-Lambertian version of SatensoRF modeling the reflective light with CP decomposition. The network is the same as module A in Fig. 6 and the final RGB is rendered with ASG functions.
3) *Lamb VM + SH:* The same with lamb CP + SH but the tensor radiance fields are composed of VM decomposition.
4) *Non-Lamb VM + ASG:* The SatensoRF itself.
5) *Non-Lamb VM + SH:* The same with non-lamb VM + ASG, but the final RGB is rendered with SH functions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

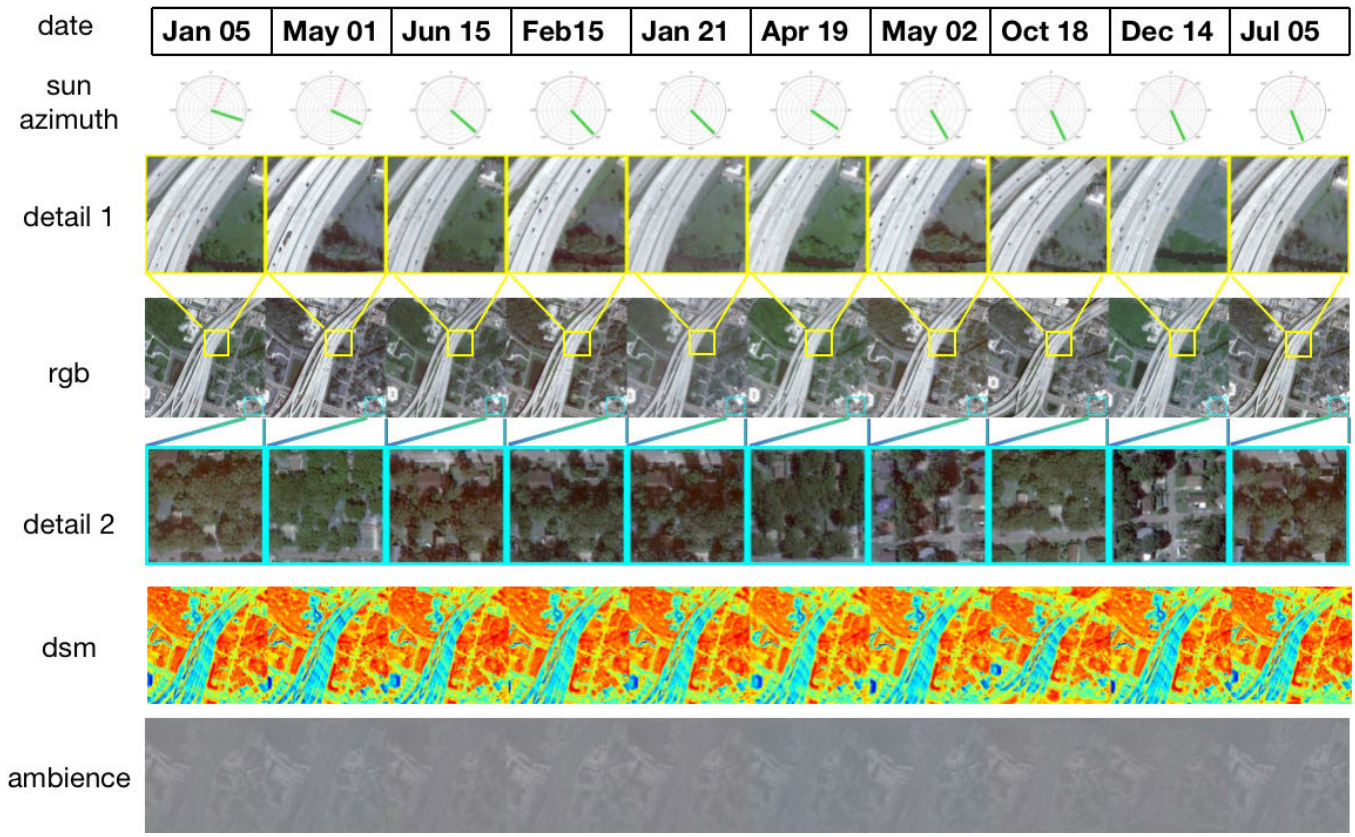


Fig. 9. Details of the synthesized images in subset 159, with asphalt viaduct details of size 500 × 500 and street details of size 300 × 300, with accurate solar direction as input.

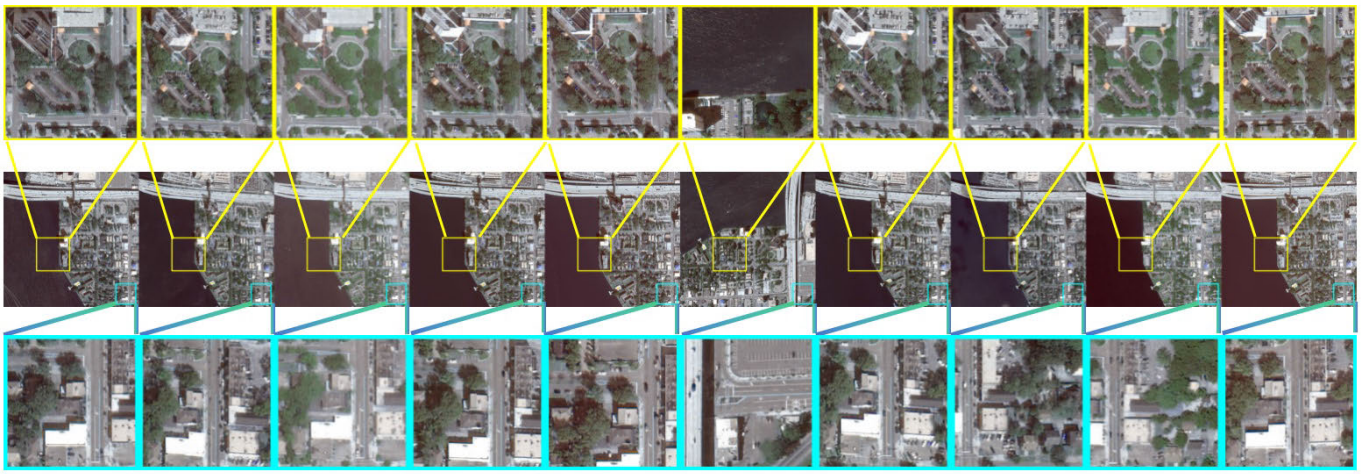


Fig. 10. Details of the synthesized images in subset 260, with water body details of size 500 × 500 and vegetation details of size 300 × 300, without accurate solar direction as input.

We evaluate the performance of these models on the test set and training set of subset 159. The quantitative results of performance and the model are listed below in Table VI, and the visualizations are summarized in Fig. 12. From Fig. 12, we can see that replacing VM decomposition with CP decomposition results in wrong volume density computation, and the wrong altitude map cannot produce DSM product correspondingly, so MAE is omitted in Table VI. When the image-based training loss converges, the volume density head converges to incorrect results when using CP decomposition, but achieves the correct convergence when utilizing VM decomposition.

E. Ablation Study II: The Total Variation Loss

This section seeks to validate the rationale behind using the total variation loss, to mitigate the adverse effects of transient objects as noise on the volume density field $\mathcal{T}_\sigma$. In the default configuration of SatensoRF, the loss function specified in (15) is employed.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

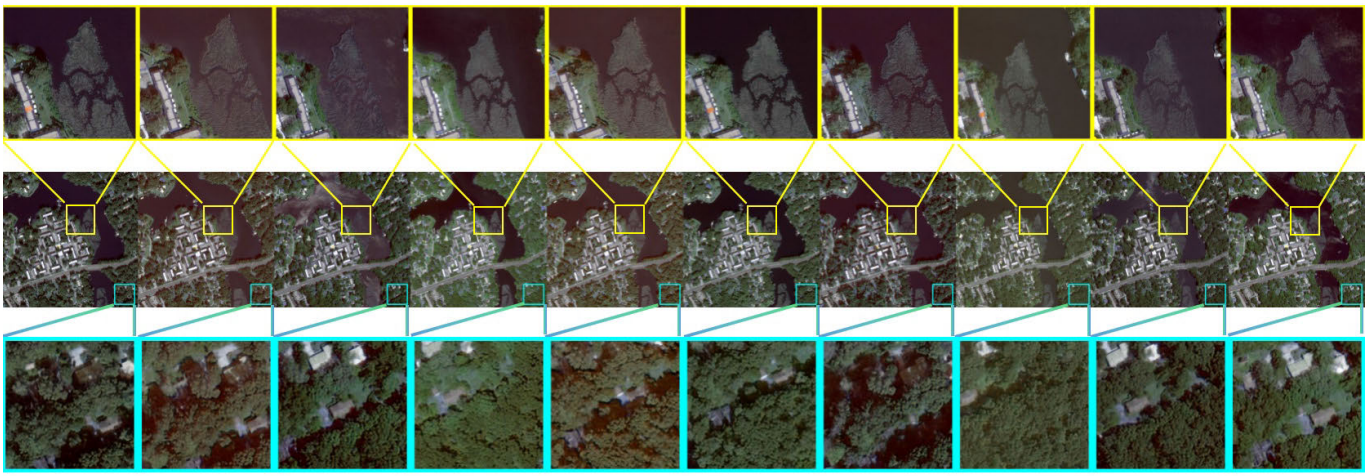


Fig. 11. Details of the synthesized images in subset 280, with water body details of size 500 × 500 and vegetation details of size 300 × 300, without accurate solar direction as input.

TABLE V

NUMERICAL RESULTS COMPARISONS OF SATENSORF AND THE STATE-OF-THE-ART WORK ON SINGLE DATE IMAGERY. POSTFIX -SC REFERS TO MODELS WITH SOLAR CORRECTION SUPERVISION IN SAT-NERF PAPER [9] AND -DS REFERS TO MODELS WITH DEPTH SUPERVISION

	PSNR(dB) ↑(test/train)				**SSIM**↑(test/train)			
Area Name	017-6	033-6	072-6	251-6	017-6	033-6	072-6	251-6
NeRF	37.012/42.561	36.984/43.784	36.971/43.659	36.969/42.998	0.901/0.923	0.982/0.988	0.913/0.968	0.948/0.952
S-NeRF + SC	36.925/45.032	35.779/43.291	36.216/42.145	35.968/40.328	0.896/0.972	0.878/0.984	0.915/0.983	0.904/0.936
Sat-NeRF + SC	36.546/43.592	37.004/44.672	36.882/44.105	36.829/43.045	0.935/0.972	0.921/0.986	0.905/0.928	0.931/0.979
SatensoRF + SC	**36.942/42.306**	**37.031/43.694**	**36.699/43.320**	**36.829/42.984**	**0.955/0.988**	**0.932/0.976**	**0.959/0.992**	**0.946/0.984**

TABLE VI

NUMERICAL RESULTS COMPARISONS OF SATENSORF WITH BRDF ASSUMPTIONS AND LAMBERTIAN ASSUMPTION

	PSNR(dB) ↑(test/train)	**SSIM**↑(test/train)	**Altitude MAE**(m)↓(test/train)	$T_{train}^{multi}(s)$	$T_{train}^{single}(s)$
lamb CP + SH	20.382/24.071	0.792/0.827	-	**8529.31**	**57.27**
non lamb CP + asg	21.518/24.599	0.781/0.847	-	10010.29	79.39
lamb VM + SH	25.473/30.074	0.872/0.901	3.104/2.945	10113.83	93.99
non lamb VM + SH	25.819/29.818	0.885/0.893	2.964/2.861	10138.73	95.85
non lamb VM + asg	**26.695/32.705**	**0.918/0.952**	**2.081/2.021**	10129.81	97.49

TABLE VII

NUMERICAL RESULTS COMPARISONS OF SATENSORF WITH TOTAL VARIATION REGULARIZATION AND $L1$ REGULARIZATION

Regularization	**PSNR**(dB) ↑(test/train)	**SSIM**↑(test/train)	**Altitude MAE**(m)↓(test/train)	$T_{train}^{multi}(s)$	$T_{train}^{single}(s)$
TV	**26.695/32.705**	**0.918/0.952**	**2.081/2.021**	10129.81	97.49
L1	26.314/31.059	0.857/0.886	-	**10096.53**	**97.29**
L1 + TV	26.518/31.622	0.872/0.894	-	10359.47	97.39

The comparative analysis encompasses the following factors.

1) *TV:* Training SatensoRF with $\mathcal{L}_{\text{TV}}$.
2) *L1:* Training SatensoRF with $\mathcal{L}_{l1}$.
3) *L1 + TV:* Training SatensoRF with both $\mathcal{L}_{l1}$ and $\mathcal{L}_{\text{TV}}$.

Where $\mathcal{L}_{l1}(x) = (1/M)\sum_{i=1}^{M-1}|\mathcal{T}_\sigma|$.

In Table VII, the validity of total variation loss $\mathcal{L}_{\text{TV}}$ is significant since it preserves the right geometry of the area, while $\mathcal{L}_{l1}(x)$ leads the model to the wrong volume density with nice rendering results. Besides, Table VII shows that once $\mathcal{L}_{l1}(x)$ is adopted, whether $\mathcal{L}_{\text{TV}}$ involves cannot guarantee the volume density estimation.

F. Controlled Experiments

SatensoRF is comprised of tensor composited fields and MLPs, and in this study, we delve into how the design of these two components influences the overall performance and efficiency of the model. For brevity, we denote model $\mathcal{T}_\sigma - m$ to be the SatensoRF model with m level of tensor radiance fields; we denote model

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
PERFORMANCE COMPARISON BY CHANGING THE TENSOR FIELD LEVEL L

	PSNR$(dB)\uparrow$	**SSIM** $\uparrow$	**Altitude MAE**(m) $\downarrow$	$T_{train}(s)$	$T_{test}(s)$	$\#Params(M)$	$FLOPs^{multi}(B)$
layer-2 $\mathcal{T}_\sigma$-2	23.208	0.877	5.382	**8189.29**	**57.23**	0.24	129.52b
layer-2 $\mathcal{T}_\sigma$-4	24.098	0.894	4.367	9101.337	70.405	0.25	132.74b
$layer$-4$\mathcal{T}_\sigma$-8	27.031	0.922	2.009	10452.052	105.31	0.86	450.57b
$layer$-2 $\mathcal{T}_\sigma$-8(SatensoRF)	**26.695**	**0.918**	**2.081**	10129.813	97.495	0.79	416.21b

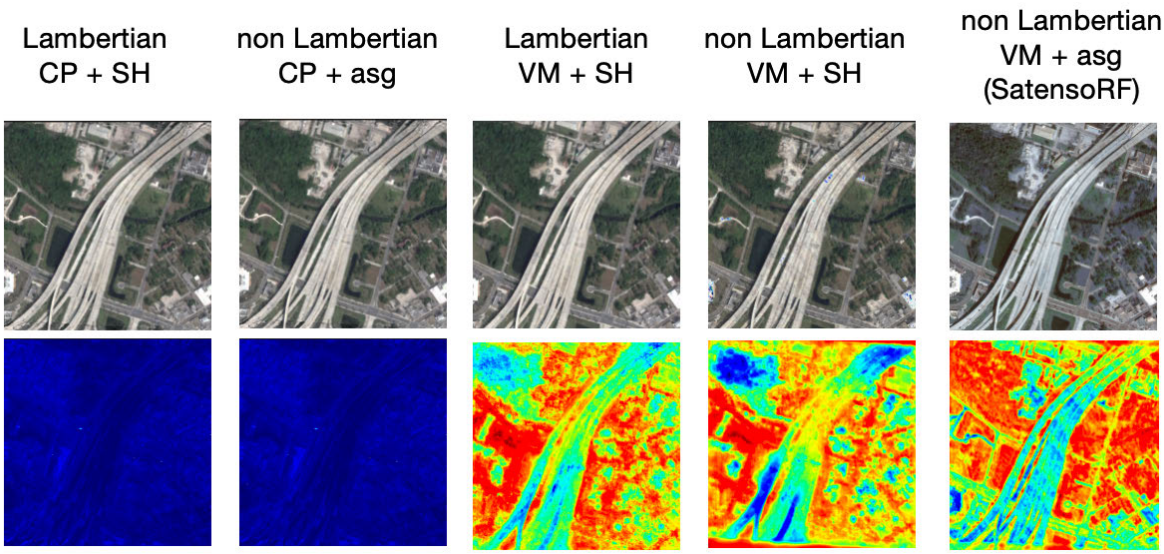


Fig. 12. Ablation study of the Lambertian assumption. A model with VM decomposition yields more valid volume density estimation, and the adoption of ASG for non-Lambertian assumption promotes model enhancement with finer details.

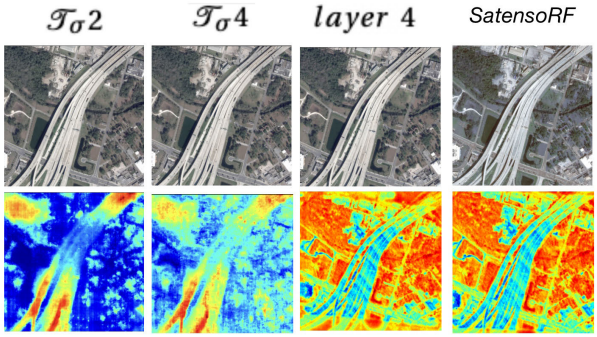


Fig. 13. Result comparison of changing tensor levels and MLP layers.

layer $-n$ to be the SatensoRF model with MLP network with n layers.

Table VIII and Fig. 13 show that both increasing the levels of tensor radiance field or the network layers enhance the performance, but the enhancement is limited while the time and computation costs are unworthy. Additionally, an insufficient level of tensor radiance field causes poor results in 3-D scene reconstruction.

V. CONCLUSION

We propose a fast, LoD tensorized NeRF called SatensoRF for large-scale satellite novel view synthesis. SatensoRF efficiently accelerates the training and inference process while improving rendering quality. To improve performance, SatensoRF additionally models the anisotropic light field instead of assuming the Earth's surface as Lambertian. Furthermore, SatensoRF does not require additional input such as solar directions. We performed a series of experiments to assess the performance and efficiency of SatensoRF. Our results indicate that SatensoRF surpasses the state-of-the-art Sat-NeRF series in terms of image rendering quality while consuming much less computation, memory, and training time on large satellite images. As a future direction, this paradigm can be extended to incorporate new dimensions such as time, enabling long-term observation of ground surfaces.

REFERENCES

[1] D. E. Shean et al., "An automated, open-source pipeline for mass production of digital elevation models (DEMs) from very-high-resolution commercial stereo satellite imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 116, pp. 101–117, Jun. 2016.

[2] G. Facciolo, C. D. Franchis, and E. Meinhardt-Llopis, "Automatic 3D reconstruction from multi-date satellite images," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 57–66.

[3] E. Rupnik, M. Pierrot-Deseilligny, and A. Delorme, "3D reconstruction from multi-view VHR-satellite images in micmac," *ISPRS J. Photogramm. Remote Sens.*, vol. 139, pp. 201–211, 2018.

[4] K. Zhang, N. Snavely, and J. Sun, "Leveraging vision reconstruction pipelines for satellite imagery," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 2139–2148.

[5] K. Gong and D. Fritsch, "DSM generation from high resolution multi-view stereo satellite imagery," *Photogrammetric Eng. Remote Sens.*, vol. 85, no. 5, pp. 379–387, May 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.

[6] J. Gao, J. Liu, and S. Ji, "A general deep learning based framework for 3D reconstruction from multi-view stereo satellite images," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 446–461, Jan. 2023.

[7] X. Wei, Y. Zhang, Z. Li, Y. Fu, and X. Xue, "DeepSFM: Structure from motion via deep bundle adjustment," in *Proc. Eur. Conf. Comput. Vis.*, Cham, Switzerland: Springer, 2020, pp. 230–247.

[8] P. Wonka, "State of the art in modeling and rendering," in *Proc. ACM SIGGRAPH Courses*, Aug. 2007, pp. 701–727.

[9] R. Marí, G. Facciolo, and T. Ehret, "Sat-NeRF: Learning multi-view satellite photogrammetry with transient objects and shadow modeling using RPC cameras," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 1310–1320.

[10] D. Maggio, M. Abate, J. Shi, C. Mario, and L. Carlone, "Loc-NeRF: Monte Carlo localization using neural radiance fields," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2023, pp. 4018–4025.

[11] H. Turki, D. Ramanan, and M. Satyanarayanan, "Mega-NeRF: Scalable construction of large-scale NeRFs for virtual fly-throughs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 12922–12931.

[12] D. Derksen and D. Izzo, "Shadow neural radiance fields for multi-view satellite photogrammetry," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 1152–1161.

[13] R. Martin-Brualla, N. Radwan, M. S. M. Sajjadi, J. T. Barron, A. Dosovitskiy, and D. Duckworth, "NeRF in the wild: Neural radiance fields for unconstrained photo collections," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7210–7219.

[14] W. Qin, J. R. Herman, and Z. Ahmad, "A fast, accurate algorithm to account for non-Lambertian surface effects on TOA radiance," *J. Geophys. Res., Atmos.*, vol. 106, no. D19, pp. 22671–22684, Oct. 2001.

[15] P. Hedman, P. P. Srinivasan, B. Mildenhall, J. T. Barron, and P. Debevec, "Baking neural radiance fields for real-time view synthesis," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 5855–5864.

[16] S. Fridovich-Keil, A. Yu, M. Tancik, Q. Chen, B. Recht, and A. Kanazawa, "Plenoxels: Radiance fields without neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5501–5510.

[17] C. Reiser et al., "MERF: Memory-efficient radiance fields for real-time view synthesis in unbounded scenes," 2023, *arXiv:2302.12249*.

[18] M. Sun et al., "Human 3D avatar modeling with implicit neural representation: A brief survey," 2023, *arXiv:2306.03576*.

[19] T. Müller, A. Evans, C. Schied, and A. Keller, "Instant neural graphics primitives with a multiresolution hash encoding," *ACM Trans. Graph.*, vol. 41, no. 4, pp. 1–15, Jul. 2022.

[20] A. Yu, R. Li, M. Tancik, H. Li, R. Ng, and A. Kanazawa, "PlenOctrees for real-time rendering of neural radiance fields," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 5752–5761.

[21] C. Sun, M. Sun, and H.-T. Chen, "Improved direct voxel grid optimization for radiance fields reconstruction," 2022, *arXiv:2206.05085*.

[22] A. Chen, Z. Xu, A. Geiger, J. Yu, and H. Su, "TensoRF: Tensorial radiance fields," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 333–350.

[23] J. Tang, X. Chen, J. Wang, and G. Zeng, "Compressible-composable nerf via rank-residual decomposition," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 14798–14809.

[24] K. Han and W. Xiang, "Multiscale tensor decomposition and rendering equation encoding for view synthesis," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 4232–4241.

[25] Y. Chang, L. Yan, B. Chen, S. Zhong, and Y. Tian, "Hyperspectral image restoration: Where does the low-rank property exist," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6869–6884, Aug. 2021.

[26] Z. Huang et al., "DLRP: Learning deep low-rank prior for remotely sensed image denoising," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[27] V. Rudnev, M. Elgharib, W. Smith, L. Liu, V. Golyanik, and C. Theobalt, "Nerf for outdoor scene relighting," in *Proc. Eur. Conf. Comput. Vis.*, Cham, Switzerland: Springer, 2022, pp. 615–631.

[28] H. Hirschmuller, "Stereo processing by semiglobal matching and mutual information," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 30, no. 2, pp. 328–341, Feb. 2008.

[29] M. Michael, J. Salmen, J. Stallkamp, and M. Schlipsing, "Real-time stereo vision: Optimizing semi-global matching," in *Proc. IEEE Intell. Vehicles Symp.*, Jun. 2013, pp. 1197–1202.

[30] K. Zhou, X. Meng, and B. Cheng, "Review of stereo matching algorithms based on deep learning," *Comput. Intell. Neurosci.*, vol. 2020, pp. 1–12, Mar. 2020.

[31] Y. Yao, Z. Luo, S. Li, T. Fang, and L. Quan, "MVSNet: Depth inference for unstructured multi-view stereo," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 767–783.

[32] X. Gu, Z. Fan, S. Zhu, Z. Dai, F. Tan, and P. Tan, "Cascade cost volume for high-resolution multi-view stereo and stereo matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2495–2504.

[33] Y. Yao, Z. Luo, S. Li, T. Shen, T. Fang, and L. Quan, "Recurrent MVSNet for high-resolution multi-view stereo depth inference," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5525–5534.

[34] K. Luo, T. Guan, L. Ju, Y. Wang, Z. Chen, and Y. Luo, "Attention-aware multi-view stereo," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1587–1596.

[35] S. Galliani, K. Lasinger, and K. Schindler, "Gipuma: Massively parallel multi-view stereo reconstruction," *Publikationen der Deutschen Gesellschaft für Photogrammetrie, Fernerkundung und Geoinformation*, vol. 25, p. 2, 2016.

[36] Q. Xu and W. Tao, "Planar prior assisted patchmatch multi-view stereo," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 12516–12523.

[37] F. Wang, S. Galliani, C. Vogel, P. Speciale, and M. Pollefeys, "PatchmatchNet: Learned multi-view patchmatch stereo," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14194–14203.

[38] K. Luo, T. Guan, L. Ju, H. Huang, and Y. Luo, "P-MVSNet: Learning patch-wise matching confidence aggregation for multi-view stereo," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 10452–10461.

[39] A. Chen et al., "MVSNeRF: Fast generalizable radiance field reconstruction from multi-view stereo," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 14124–14133.

[40] Y. Wei, S. Liu, Y. Rao, W. Zhao, J. Lu, and J. Zhou, "NerfingMVS: Guided optimization of neural radiance fields for indoor multi-view stereo," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 5610–5619.

[41] K. Deng, A. Liu, J.-Y. Zhu, and D. Ramanan, "Depth-supervised NeRF: Fewer views and faster training for free," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 12882–12891.

[42] J. Chibane, A. Bansal, V. Lazova, and G. Pons-Moll, "Stereo radiance fields (SRF): Learning view synthesis for sparse views of novel scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7911–7920.

[43] B. Kaya, S. Kumar, F. Sarno, V. Ferrari, and L. Van Gool, "Neural radiance fields approach to deep multi-view photometric stereo," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 3967–3979.

[44] J. T. Kajiya, "The rendering equation," in *Proc. 13th Annu. Conf. Comput. Graph. Interact. Techn.*, Aug. 1986, pp. 143–150.

[45] M. Pharr, W. Jakob, and G. Humphreys, *Physically Based Rendering: From Theory to Implementation*. Cambridge, MA, USA: MIT Press, 2023.

[46] H. Jin et al., "TensoIR: Tensorial inverse rendering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 165–174.

[47] K. Xu, W.-L. Sun, Z. Dong, D.-Y. Zhao, R.-D. Wu, and S.-M. Hu, "Anisotropic spherical Gaussians," *ACM Trans. Graph.*, vol. 32, no. 6, pp. 1–11, Nov. 2013.

[48] Z. Lin, "A review on low-rank models in data analysis," *Big Data Inf. Analytics*, vol. 1, no. 2/3, pp. 139–161, Aug. 2016.

[49] Y. Chang, L. Yan, H. Fang, and C. Luo, "Anisotropic spectral–spatial total variation model for multispectral remote sensing image destriping," *IEEE Trans. Image Process.*, vol. 24, no. 6, pp. 1852–1866, Jun. 2015.

[50] W. He, N. Yokoya, L. Yuan, and Q. Zhao, "Remote sensing image reconstruction using tensor ring completion and total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8998–9009, Nov. 2019.

[51] D. Verbin, P. Hedman, B. Mildenhall, T. Zickler, J. T. Barron, and P. P. Srinivasan, "Ref-NeRF: Structured view-dependent appearance for neural radiance fields," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5481–5490.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.



Tongtong Zhang received the B.S. degree from the School of Mathematics and Statistics, Xidian University, Xi'an, China, in 2019.

She is currently pursuing the Ph.D. degree with the School of Aeronautics and Astronautics, Shanghai Jiao Tong University, Shanghai, China. Her major research interests include 3-D reconstruction, neural radiance field, camera bundle adjustment, and calibration.



Yu Zhou received the Ph.D. degree in surveying and mapping engineering from the Information Engineering University, Zhengzhou, China, in 2020.

Since 2016, he has been a Research Assistant with the Xi'an Institute of Surveying and Mapping, Xi'an, China. He has designed and developed the operational attitude postprocessing software and ground data-processing system of the Gaofen-14 satellite. His main research interests include technology of optical cartographic satellite attitude postprocessing, in-orbit geometry calibration, and optical image 3-D reconstruction.



Yuanxiang Li (Member, IEEE) received the Ph.D. degree in signal and information processing from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2001.

From 2002 to 2004, he was a Research Fellow with the School of Computing, National University of Singapore, Singapore. From 2015 to 2016, he was a Visiting Professor with the University of Michigan, MI, USA. He is currently a Professor with the School of Aeronautics and Astronautics, Shanghai Jiao Tong University, Shanghai, China. His research interests include machine learning, default diagnosis and prediction, time-series signal analysis, data and knowledge fusion, image recognition, and 3-D reconstruction.



Xian Wei (Senior Member, IEEE) received the Ph.D. degree in computer engineering from the Technical University of Munich, Munich, Germany, in 2017.

He is currently a Research Professor with the Software Engineering Institute (SEI), East China Normal University, Shanghai, China. He has authored over 100 publications in refereed journals and conference proceedings. His research interests focus on sparse coding, deep learning, geometric optimization, and time-series analysis. The applications include multisensor fusion for intelligent cars, robotic vision, data sequence or image modeling, synthesis, recognition, and semantics.

Dr. Wei is a Senior Member of the CCF.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:36:35 UTC from IEEE Xplore. Restrictions apply.